

①

Exercice :1) la question : vérifier que $\bar{x} = 82,5$ et $s = \hat{s} = 6,86$

$$\text{Réponse: } \frac{n=4}{x} = \bar{x}_4 = \frac{1}{4} \sum_{k=1}^4 x_k = \frac{1}{4} (78 + 85 + 91 + 76) = 82,5$$

$$s^2 = \hat{s}^2 = \frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x}_n)^2 \quad (n=4)$$

$$= \frac{1}{3} \sum_{k=1}^4 (x_k - 82,5)^2 = 47.$$

$$\text{Donc } s = \hat{s} = \sqrt{47} \approx 6,86.$$

2) Dans le cas où σ est connu (donné $\sigma = 5$)

$$n=4; \quad \alpha = 5\% = 0,05; \quad 1-\alpha = 0,95$$

$$z_{1-\alpha/2} = z_{0,975} = F^{-1}_{(0,975)} = 1,96$$

$$\text{et IC}_{1-\alpha}(x) = \left[\bar{x}_n - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}; \bar{x}_n + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

$$\text{et IC}_{0,95}(x) = [77,6; 87,4].$$

3) σ est estimé empirique; $s = \hat{s} = 6,86$

$$n=4; \quad \text{loi de Student } \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{s^2} \approx \chi^2_3$$

$$t_{1-\alpha/2} = t_{0,975} = F^{-1}_{(0,975)} = 3,182$$

$$\text{et IC}_{1-\alpha}(x) = \left[\bar{x}_n - t_{1-\alpha/2} \frac{\hat{s}}{\sqrt{n}}; \bar{x}_n + t_{1-\alpha/2} \frac{\hat{s}}{\sqrt{n}} \right]$$

$$= [71,6; 93,4].$$